

**Executive Summary:** This week the team got a lot done in terms of future plans for the project. On the hardware side of things, every schematic has been completed and a final combined schematic is completed. This along with every schematic being uploaded to Github means that the hardware team is ready for building the moment all parts are in. On the software side of things we got closer to finishing the bluetooth adaptation and audio sampling and the GPS software was started this week. Our team also submitted our parts list for ordering with slight issues on sites ordering from, however, after slight alterations all parts are ordered and should be ready for building after spring break. Our team is taking a small break over spring break where progress will be made on our CDR presentation but other than this, there will be less progress on the actual project itself.

Hardware:

Cody C.

Progress:

- 3/11 – Meeting with Ryan to collaborate on schematics (~1 Hour)
- Final touches on the combined schematics (~1 Hour)
  - Updated library in KiCad
  - Created two schematics: One with labels and one with wiring
- Upload all schematics to GitHub
- Updated the “Gunshot Detection Files” folder for organization purposes (~10 Minutes)

**Total Hours Worked:** ~ 2 Hours 10 Minutes

Future Work/Plan

- Gather information for CDR
- Work on presentation

- Practice presentation if completed with slides
- Create talk bullet points if completed with slides

Hardware:

**Ryan K.**

Progress:

- 3/11 (1hr)
  - Finalized schematics with Cody
- Downloaded all KiCad symbols and schematics for future PCB layout design (~25min)

**Total Hours Worked:** ~ 1hr 25min

Future Work/Plan:

- Work on CDR slides for upcoming presentation on 3/30

Software:

**Alex H.**

Progress:

- 3/11 (27 min)
  - Continued troubleshooting Bluetooth connection code
  - Downloaded nRF Connect mobile phone app to receive serial messages from Wearable Alert Devices. Since the mobile app for the Wearable Alert Device is not functional, nRF Connect will be used to simulate a mobile phone during the final demonstration.
- 3/13 (3 h 46 min)

- Fixed software bugs with Bluetooth connectivity functionality. The issue was that the software did not give the Wearable Alert Device a NOTIFY property, which did not allow automatic updates of incoming messages on the nRF Connect mobile app.
- Created fake data to test wireless data transmission.
- **nRF Connect mobile app successfully received messages from Wearable Alert Device. Wireless data transmission functionality for the project has been completed.**
- 3/14 (4 h 53 min)
  - Begun work on GPS Module software development.
  - Wrote I2C transmission code for software, reading datasheets for the GPS and RTC module for register addresses.
  - Later realized that the GPS and RTC module chip came with its own software library, since the chip made its own custom I2C addresses to combine the GPS and RTC modules together into a single board.
  - Downloaded software library for GPS module (DFRobot\_GNSSAndRTC.h/.cpp)
  - Wrote software for collecting timestamp from GPS module.

**Total Hours Worked:** 9 h 6 min

Future Work/Plan:

- Begin working on audio sampling and signal processing software for Gunshot Detection
- Create a C struct that represents a data packet for the Wearable Alert Device. Need to discuss with the team about what information should be put in this data packet.

Project Manager:

**Jonas W.**

Progress:

- 3/8 (30 minutes)
  - Format and begin the process of ordering parts.
- 3/10 (1 Hr)
  - Complete the list of items and email to Dr. Fitzmorris. Parts getting ordered
- 3/12 (15 Minutes)
  - Complete the issue of some of the parts ordered and get them resubmitted
- 3/13 (15 Minutes)
  - Run by the ECE office to get the first round of items

**Total Hours Worked:** ~ 2 Hours

Future Work/Plan:

Next week I plan on beginning the CDR and working on it although it is spring break. The other parts should arrive as well even though it is spring break and I will be able to pick those up after spring break as well.